# Final Project

## STAT 311 Regression Analysis

Spring 2026

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# An Analysis of the Effect of Social Media Use on Productivity

## Introduction

Social media use is increasingly cited as a factor in declining productivity, but the conversation tends to lump all screen time together. This project looks at whether social media specifically has a stronger negative effect on productivity than total screen time alone, using survey data from 6,000 respondents collected by an independent researcher and posted to Kaggle. The dataset includes self-reported measures of daily screen time, daily social media use, study hours, sleep hours, daily notifications received, a focus score, an ordinal addiction level (Low, Medium, High), and a productivity score on a 0 to 100 scale.

The main research question is whether social media hours, as a distinct predictor, contributes to a productivity model beyond what general screen time can explain. If the two were essentially measuring the same thing, we would expect them to behave similarly in a regression model. If social media has its own effect, it should remain a significant predictor even when daily screen time is included alongside it. The analysis also looks at how other lifestyle variables such as study hours, sleep, and notification frequency interact with social media use to influence productivity.

The approach is a standard multiple regression workflow. We start with exploratory data analysis to understand variable distributions and check for outliers. We then build a sequence of models, beginning with a first-order quantitative model and progressively adding the categorical addiction level, removing variables that stepwise selection identifies as non-contributing, and finally testing higher-order terms and interactions justified by partial F-tests. The goal is the simplest model that adequately explains the data, with all assumptions reasonably satisfied. The final model is also validated on a held-out portion of the data to check that the fit generalizes.

## Data Description

The dataset was downloaded from Kaggle and contains 6,000 observations across nine variables. The data was generated through a survey instrument capturing self-reported behaviors and outcomes. Respondents reported their average daily screen time in hours, the portion of that screen time spent on social media, hours spent studying or doing focused work, hours of sleep, the number of notifications received per day, a self-rated focus score on a 0 to 100 scale, an ordinal self-classification of addiction level (Low, Medium, or High), and a productivity score on a 0 to 100 scale, which serves as the response variable. Age was also recorded. The dataset was uploaded by a single user without a published survey methodology or sampling frame, and analyses below should be interpreted with this in mind.

The dataset has missing values scattered across columns. Each individual column has 120 missing values, but the missingness is not concentrated in the same rows; in total, 1,001 of the 6,000 rows (about 16.7 percent) contain at least one missing value. Of these, 927 rows have a

single missing value, 69 have two, and 5 have three. Because there is no obvious pattern to the missingness and listwise deletion still leaves a large sample, we drop any row with at least one missing value. This leaves 4,999 complete observations going into Model 1.

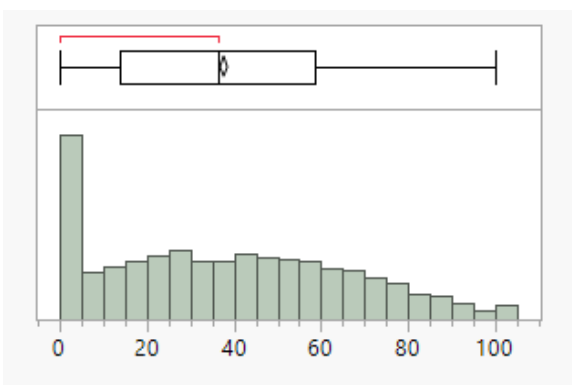
Variables and roles in the analysis:

- productivity_score: continuous response variable, range 0 to 100
- age: continuous predictor, range 15 to 39
- daily_screen_time: continuous predictor, total daily screen time in hours
- social_media_hours: continuous predictor, daily hours on social media
- study_hours: continuous predictor, daily hours of focused work or study
- sleep_hours: continuous predictor, hours of sleep per night
- notifications_per_day: continuous predictor, count of notifications received daily
- focus_score: continuous predictor, self-reported focus rating 0 to 100
- addiction_level: categorical predictor, three levels (Low, Medium, High)

All predictors are individual-level survey responses with no temporal or spatial structure, satisfying the independence assumption discussed in class.

## Exploratory Data Analysis

We start by looking at the distribution of productivity_score, the response variable, using a histogram and box plot shown in Figure 1 along with the summary statistics in Figure 2.



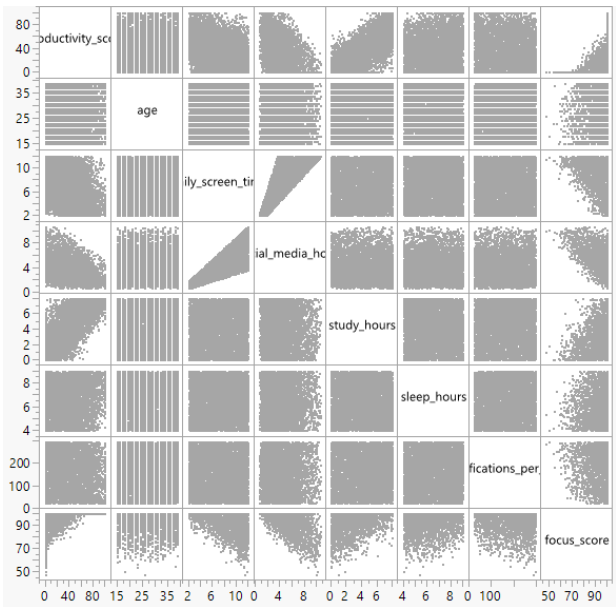

**Summary Statistics**

| | |
|---|---|
| Mean | 37.743095 |
| Std Dev | 27.627953 |
| Std Err Mean | 0.3907573 |
| Upper 95% Mean | 38.50915 |
| Lower 95% Mean | 36.977039 |
| N | 4999 |
| N Missing | 0 |

*Figure 1: Histogram and box plot of productivity_score.* *Figure 2: Summary statistics.*

The distribution is roughly uniform across the 0 to 100 range with a slight right skew. The mean is 37.74, the median is 36.46, and the standard deviation is 27.63. The minimum is 0 and the maximum is 100. One feature worth flagging is the clustering at the boundaries; about 14 percent of respondents scored exactly 0 and another small share scored exactly 100. This may reflect genuine responses at the extremes or an artifact of how the data was generated, and it is something we revisit when interpreting residual diagnostics later.

Before building any models, we examined the distributions and relationships of each predictor with the response. The scatterplot matrix in Figure 3 shows productivity_score plotted against each quantitative predictor, with regression lines overlaid in Figure 4.



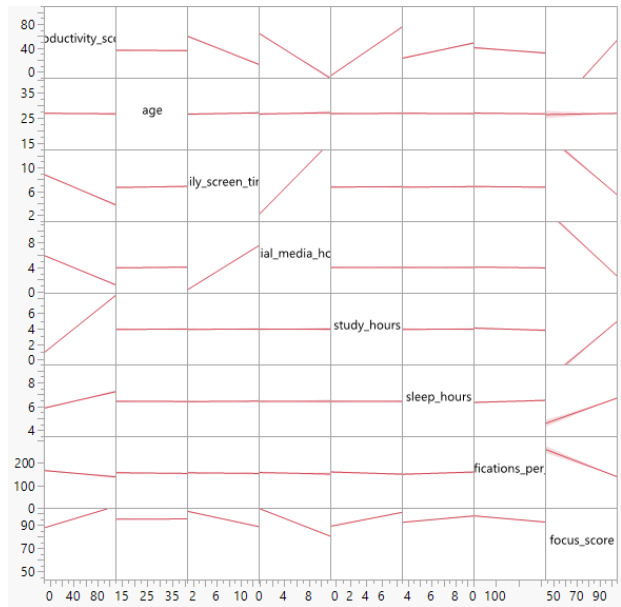




*Figure 3: Scatterplot matrix.* *Figure 4: Same matrix with linear fit lines.*

Several patterns are visible. social_media_hours shows a clear negative linear relationship with productivity, and daily_screen_time shows a similar but somewhat weaker negative trend. study_hours has the strongest positive relationship, and sleep_hours shows a moderate positive trend. focus_score has an unusual distribution; most respondents report focus scores at or near 100, with a smaller cluster scattered between 50 and 95. This left-skew is worth keeping in mind for residual diagnostics later. age and notifications_per_day show no strong univariate trend with productivity.

Pearson correlations between each predictor and productivity_score are shown in Figure 5.

| | productivity_score | age | daily_screen_time | social_media_hours | study_hours | sleep_hours | notifications_per_day | focus_score |
|---|---|---|---|---|---|---|---|---|
| productivity_score | 1.0000 | -0.0074 | -0.4093 | -0.5075 | 0.7383 | 0.2207 | -0.0820 | 0.5747 |
| age | -0.0074 | 1.0000 | 0.0163 | 0.0138 | 0.0059 | -0.0048 | -0.0103 | 0.0079 |
| daily_screen_time | -0.4093 | 0.0163 | 1.0000 | 0.7931 | 0.0055 | 0.0067 | -0.0087 | -0.4261 |
| social_media_hours | -0.5075 | 0.0138 | 0.7931 | 1.0000 | 0.0021 | 0.0026 | -0.0153 | -0.6005 |
| study_hours | 0.7383 | 0.0059 | 0.0055 | 0.0021 | 1.0000 | 0.0021 | -0.0305 | 0.4030 |
| sleep_hours | 0.2207 | -0.0048 | 0.0067 | 0.0026 | 0.0021 | 1.0000 | 0.0296 | 0.1780 |
| notifications_per_day | -0.0820 | -0.0103 | -0.0087 | -0.0153 | -0.0305 | 0.0296 | 1.0000 | -0.1820 |
| focus_score | 0.5747 | 0.0079 | -0.4261 | -0.6005 | 0.4030 | 0.1780 | -0.1820 | 1.0000 |

*Figure 5: Correlation matrix among quantitative variables.*

The correlations confirm the visual patterns. study_hours has the strongest positive correlation with productivity at 0.738. focus_score is next at 0.575. social_media_hours has a strong negative correlation at -0.508, and daily_screen_time is also negative at -0.409. sleep_hours is

positively correlated at 0.221, while age and notifications_per_day have weak correlations at -0.007 and -0.082 respectively. Importantly, daily_screen_time and social_media_hours are correlated with each other at 0.793, which signals possible multicollinearity that we will need to keep in mind during model building.

The categorical variable addiction_level has three levels with the following approximate counts in the cleaned data: Low (about 16 percent), Medium (about 52 percent), and High (about 32 percent). A box plot of productivity_score by addiction_level is shown in Figure 6.

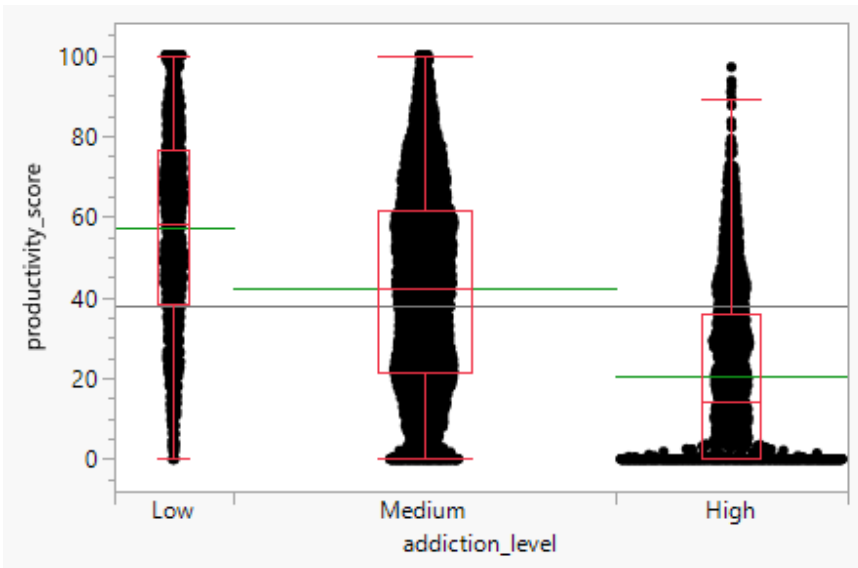



*Figure 6: Productivity score by addiction level.*

There is a visible difference in median productivity across the three groups. Mean productivity drops stepwise across the groups: about 58 for Low addiction, 43 for Medium, and 22 for High. Whether this difference holds up after controlling for the quantitative predictors is something we test directly in Model 2.

## Methods

The analysis follows a sequential model-building strategy. We begin with Model 1, a first-order linear model using only the seven quantitative predictors. The purpose of this model is twofold: it gives us a baseline for comparing future models, and it produces influence diagnostics (Cook's Distance, externally studentized residuals, hat values) that we use to identify and exclude any unduly influential observations. After excluding these observations, we refit Model 1 on the cleaned data.

Model 2 adds the categorical predictor addiction_level. JMP handles categorical predictors automatically, producing parameter estimates for each non-reference level. This allows us to evaluate whether self-reported addiction level adds explanatory power on top of the quantitative measurements.

We then apply stepwise selection methods (All Possible Models, Forward, Backward, and Mixed Stepwise) using a p-value threshold of 0.05 to enter and leave. The point of running multiple methods is to see whether they converge on the same subset of variables; consistent results across methods give us more confidence in the variable selection. We use AIC, BIC, Adjusted R-squared, and Mallows' Cp as comparison metrics. The reduced model selected by stepwise becomes Model 3.

Model 4 introduces second-order terms for any predictors where adding a quadratic term substantially improves fit. Each addition is justified with a partial F-test comparing the nested models. Model 5 adds interaction terms motivated by the research question and by the EDA, again validated with partial F-tests. Following standard practice, we restricted second-order and interaction term testing to variables retained by stepwise selection, in keeping with the principle of marginality.

Throughout the process, residual diagnostics are checked at each stage: residual versus predicted plots for patterns or heteroscedasticity, normal quantile plots for normality of residuals, and externally studentized residuals for outliers. The final model is selected on the basis of fit (Adjusted R-squared and MSE), parsimony (preferring fewer terms when fit is comparable), and assumption compliance. As a final check on generalizability, the final model is fit on a random 80 percent of the data and used to predict the remaining 20 percent.

## Analysis

### Model 1: First-Order Quantitative Predictors

Model 1 uses all seven quantitative predictors with no transformations or interactions:

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7$$

Where:

- $x_1$ = age
- $x_2$ = daily_screen_time
- $x_3$ = social_media_hours
- $x_4$ = study_hours
- $x_5$ = sleep_hours
- $x_6$ = notifications_per_day
- $x_7$ = focus_score

**Hypothesis:**

$$H_0: \beta_1 = \beta_2 = \ldots = \beta_7 = 0 \quad \text{vs.} \quad H_1: \text{at least one } \beta_i \neq 0$$

**Output:**

| Summary of Fit | |
|---|---|
| RSquare | 0.874627 |
| RSquare Adj | 0.874451 |
| Root Mean Square Error | 9.78939 |
| Mean of Response | 37.74309 |
| Observations (or Sum Wgts) | 4999 |

**Analysis of Variance**

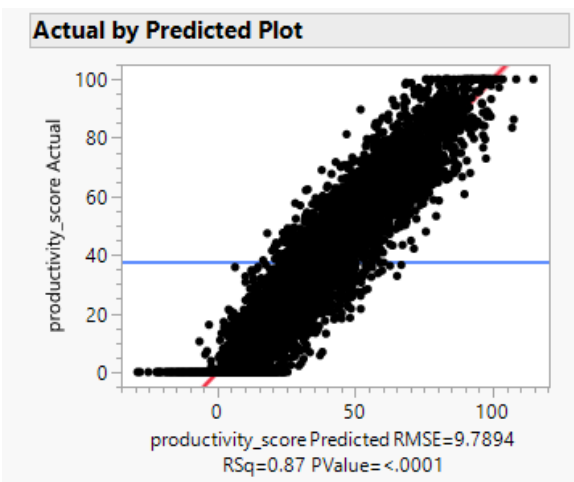
| Source | DF | Sum of Squares | Mean Square | F Ratio |
|---|---|---|---|---|
| Model | 7 | 3336694.0 | 476671 | 4974.015 |
| Error | 4991 | 478298.3 | 96 | Prob > F |
| C. Total | 4998 | 3814992.2 | | <.0001* |

*Figure 7: Model 1 Summary of Fit and Analysis of Variance.*

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| |
|---|---|---|---|---|
| Intercept | 76.678489 | 2.948451 | 26.01 | <.0001* |
| age | -0.006727 | 0.019078 | -0.35 | 0.7244 |
| daily_screen_time | -0.020393 | 0.080075 | -0.25 | 0.7990 |
| social_media_hours | -8.0596 | 0.127318 | -63.30 | <.0001* |
| study_hours | 9.7467637 | 0.070641 | 137.98 | <.0001* |
| sleep_hours | 4.9943055 | 0.100503 | 49.69 | <.0001* |
| notifications_per_day | -0.03767 | 0.001777 | -21.20 | <.0001* |
| focus_score | -0.741419 | 0.029327 | -25.28 | <.0001* |

*Figure 8: Model 1 Parameter Estimates.*



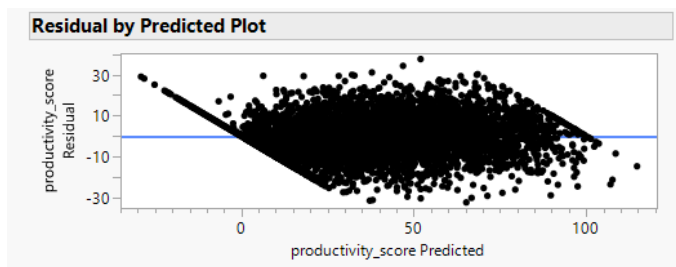




*Figure 9: Model 1 Actual by Predicted plot.* *Figure 10: Residual by Predicted plot.*

Model 1 produces an R-squared of 0.8746 and an Adjusted R-squared of 0.8745, meaning the model accounts for about 87.5 percent of the variance in productivity_score. The F-statistic is 4974.0 with a p-value below 0.0001, so we reject the null hypothesis; the model is statistically significant overall. The Root Mean Square Error is 9.79.

Looking at the individual parameter estimates, five of the seven predictors are highly significant ($p < 0.0001$): social_media_hours (estimate -8.06), study_hours (estimate 9.75), sleep_hours

(estimate 4.99), notifications_per_day (estimate -0.038), and focus_score (estimate -0.74). Two predictors are not statistically significant: age (p = 0.7244) and daily_screen_time (p = 0.7990). The non-significance of daily_screen_time once social_media_hours is in the model is the first piece of evidence directly addressing our research question; total screen time, after accounting for social media specifically, does not add useful information.

To formally confirm this, we look at the Effect Tests output for Model 1, shown in Figure 11. The Effect Test for daily_screen_time gives F = 0.065 with p = 0.7990. This is the partial F-test for whether daily_screen_time adds explanatory power to a model that already contains social_media_hours and the other predictors; we fail to reject the null hypothesis that its coefficient is zero. In other words, after controlling for social media use, raw daily screen time provides no additional information about productivity. This directly answers the research question.

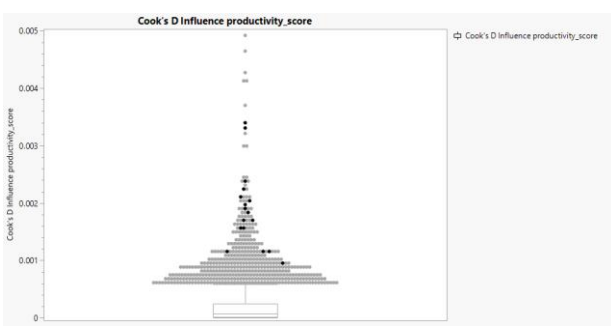
**Effect Tests**

| Source | Nparm | DF | Sum of Squares | F Ratio | Prob > F |
|---|---|---|---|---|---|
| age | 1 | 1 | 11.9 | 0.1243 | 0.7244 |
| daily_screen_time | 1 | 1 | 6.2 | 0.0649 | 0.7990 |
| social_media_hours | 1 | 1 | 384024.1 | 4007.257 | <.0001* |
| study_hours | 1 | 1 | 1824388.4 | 19037.33 | <.0001* |
| sleep_hours | 1 | 1 | 236646.6 | 2469.386 | <.0001* |
| notifications_per_day | 1 | 1 | 43059.9 | 449.3264 | <.0001* |
| focus_score | 1 | 1 | 61250.0 | 639.1385 | <.0001* |

*Figure 11: Effect Tests for Model 1.*

The negative coefficient on focus_score is initially counterintuitive but makes sense given the EDA finding that focus_score is heavily clustered near 100 with a small tail of lower values. The variable is interacting with the other predictors in a way we will revisit when we consider higher-order terms.

**Influence Diagnostics:**



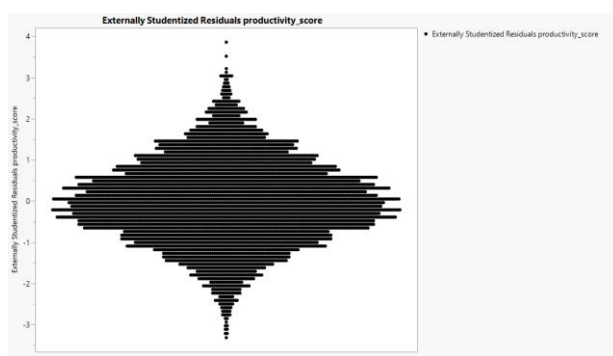




*Figure 12: Cook's Distance.* *Figure 13: Externally Studentized Residuals.*

The maximum Cook's Distance is approximately 0.005, well below the threshold of 1.0 typically used to flag influential observations. No observations exceed this threshold. However, 17 observations have absolute externally studentized residuals greater than 3, which is the standard cutoff used to flag potential outliers. Hat values flag some observations as having high leverage, which is expected in any large dataset and is not by itself a reason for exclusion.

We exclude the 17 observations with absolute studentized residuals greater than 3 and refit Model 1 on the remaining 4,982 rows. The refit yields a slight improvement: R-squared rises to 0.8783, Adjusted R-squared to 0.8781, and RMSE drops to 9.64. All subsequent models use this cleaned dataset of 4,982 observations.

**Summary of Fit**

| | |
|---|---|
| RSquare | 0.878271 |
| RSquare Adj | 0.8781 |
| Root Mean Square Error | 9.636724 |
| Mean of Response | 37.68774 |
| Observations (or Sum Wgts) | 4982 |

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| |
|---|---|---|---|---|
| Intercept | 76.861378 | 2.905162 | 26.46 | <.0001* |
| age | -0.003104 | 0.018818 | -0.16 | 0.8690 |
| daily_screen_time | -0.026056 | 0.078882 | -0.33 | 0.7412 |
| social_media_hours | -8.045216 | 0.125515 | -64.10 | <.0001* |
| study_hours | 9.7690104 | 0.069666 | 140.23 | <.0001* |
| sleep_hours | 4.983394 | 0.099131 | 50.27 | <.0001* |
| notifications_per_day | -0.037574 | 0.001753 | -21.44 | <.0001* |
| focus_score | -0.745241 | 0.028894 | -25.79 | <.0001* |

*Figure 14: Refit Model 1 Summary of Fit. Figure 15: Refit Parameter Estimates.*

## Model 2: Adding addiction_level

Model 2 adds the categorical predictor addiction_level. JMP automatically handled the encoding by treating High as the reference category, with parameter estimates produced for the Low and Medium effects.

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \text{(effect terms for addiction_level)}$$

**Hypothesis:**

$H_0$: all coefficients including addiction_level effects = 0 vs. $H_1$: at least one ≠ 0

**Output:**

**Summary of Fit**

| | |
|---|---|
| RSquare | 0.87829 |
| RSquare Adj | 0.87807 |
| Root Mean Square Error | 9.637909 |
| Mean of Response | 37.68774 |
| Observations (or Sum Wgts) | 4982 |

**Analysis of Variance**

| Source | DF | Sum of Squares | Mean Square | F Ratio |
|---|---|---|---|---|
| Model | 9 | 3332797.7 | 370311 | 3986.583 |
| Error | 4972 | 461845.6 | 93 | Prob > F |
| C. Total | 4981 | 3794643.3 | | <.0001* |

*Figure 16: Model 2 Summary of Fit and Analysis of Variance.*

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| |
|---|---|---|---|---|
| Intercept | 76.564431 | 3.055271 | 25.06 | <.0001* |
| age | -0.002538 | 0.018831 | -0.13 | 0.8928 |
| daily_screen_time | -0.034249 | 0.081319 | -0.42 | 0.6736 |
| social_media_hours | -7.9546 | 0.178081 | -44.67 | <.0001* |
| study_hours | 9.7691999 | 0.070346 | 138.87 | <.0001* |
| sleep_hours | 4.9832704 | 0.099421 | 50.12 | <.0001* |
| notifications_per_day | -0.037582 | 0.001754 | -21.42 | <.0001* |
| focus_score | -0.745705 | 0.0296 | -25.19 | <.0001* |
| addiction_level[Low] | 0.1510137 | 0.425452 | 0.35 | 0.7226 |
| addiction_level[Medium] | 0.157629 | 0.206523 | 0.76 | 0.4454 |

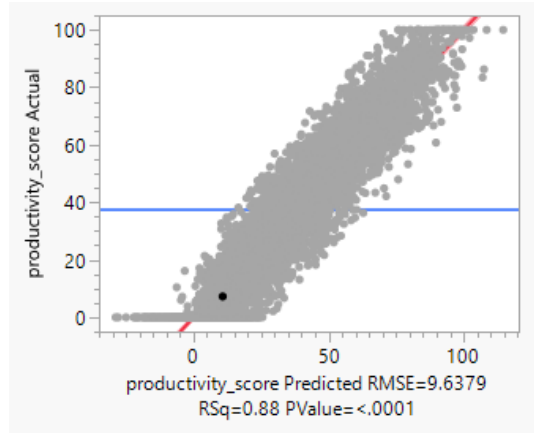



*Figure 17: Model 2 Parameter Estimates.* *Figure 18: Actual by Predicted plot.*

Model 2 yields R-squared 0.8783 and Adjusted R-squared 0.8781, essentially identical to refitted Model 1. The F-statistic is 3986.6, still highly significant. With High as the reference category, neither the Low effect ($p = 0.7226$) nor the Medium effect ($p = 0.4454$) is statistically significant. The MSE is essentially unchanged from refitted Model 1.

This is an informative result. Self-reported addiction level does not add explanatory power once we control for actual hours spent on social media, screen time, and the other quantitative variables. The information that someone classifies themselves at a particular addiction level is essentially captured by their reported social media hours and other behavioral measures. We will allow stepwise selection to confirm whether to drop addiction_level from further modeling.

## Stepwise Selection

We ran All Possible Models, Forward, Backward, and Mixed Stepwise procedures in JMP using a p-value threshold of 0.05. The All Possible Models procedure identifies the optimal model by AICc and Cp, and the result is shown in Figure 19, with the corresponding RMSE-by-model-size visualization in Figure 20.

| SSE | DFE | RMSE | RSquare | RSquare Adj | Cp | p | AICc | BIC |
|---|---|---|---|---|---|---|---|---|
| 461930.45 | 4976 | 9.6349199 | 0.8783 | 0.8781 | 2.9139505 | 6 | 36718.71 | 36764.28 |

*Figure 19: Top model from All Possible Models procedure (lowest AICc and Cp).*

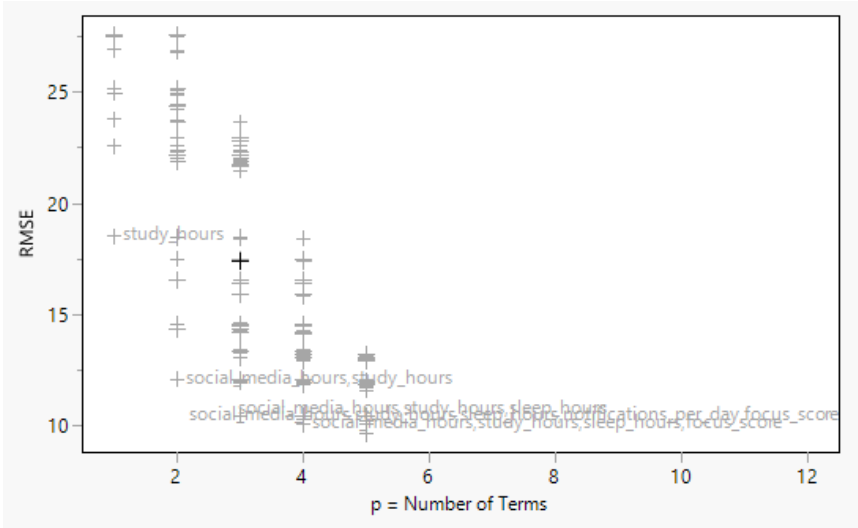



*Figure 20: RMSE versus number of model terms.*

The top model has 5 predictors, AICc 36,718.71, BIC 36,764.28, Cp 2.91, and Adjusted R-squared 0.8781. The RMSE-by-terms plot shows that fit improves rapidly as the right variables enter and then plateaus once social_media_hours, study_hours, sleep_hours, notifications_per_day, and focus_score are included. Adding more variables beyond those five does not meaningfully improve RMSE.

The Forward, Backward, and Mixed Stepwise procedures all converged on the same five-variable model. Figure 21 shows the final Current Estimates table, with the five retained predictors marked Entered. age, daily_screen_time, and the addiction_level effect terms all failed to enter or were removed during the procedure.

**Current Estimates**

| Lock | Entered | Parameter | Estimate | nDF | SS | "F Ratio" | "Prob>F" |
|---|---|---|---|---|---|---|---|
| ☑ | ☑ | Intercept | 76.8358472 | 1 | 0 | 0.000 | 1 |
| ☐ | ☐ | age | 0 | 1 | 2.582964 | 0.028 | 0.86754 |
| ☐ | ☐ | daily_screen_time | 0 | 1 | 10.18895 | 0.110 | 0.74046 |
| ☐ | ☑ | social_media_hours | -8.0754323 | 1 | 799816.5 | 8615.771 | 0 |
| ☐ | ☑ | study_hours | 9.77045743 | 1 | 1833323 | 19748.89 | 0 |
| ☐ | ☑ | sleep_hours | 4.98444411 | 1 | 234985.2 | 2531.304 | 0 |
| ☐ | ☑ | notifications_per_day | -0.0375947 | 1 | 42785.15 | 460.890 | 7.5e-98 |
| ☐ | ☑ | focus_score | -0.7465235 | 1 | 62986.92 | 678.507 | 2e-140 |
| ☐ | ☐ | addiction_level{High-Medium&Low} | 0 | 1 | 66.04628 | 0.711 | 0.39901 |
| ☐ | ☐ | addiction_level{Medium-Low} | 0 | 1 | 0.592135 | 0.006 | 0.93635 |

*Figure 21: Final Stepwise selection result; checked variables are retained.*

The convergence across multiple stepwise methods, combined with the All Possible Models result, gives strong confidence in the variable selection. Variables that did not survive stepwise are exactly those flagged as either non-significant in Model 1 (age, daily_screen_time) or non-significant in Model 2 (addiction_level).

## Model 3: Reduced Model from Stepwise

Model 3 uses the five variables retained by stepwise selection.

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5$$

Where:

- $x_1$ = social_media_hours
- $x_2$ = study_hours
- $x_3$ = sleep_hours
- $x_4$ = notifications_per_day
- $x_5$ = focus_score

**Hypothesis:**

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0 \quad \text{vs.} \quad H_1: \text{at least one } \beta_i \neq 0$$

**Output:**

**Summary of Fit**

| | |
|---|---|
| RSquare | 0.878268 |
| RSquare Adj | 0.878145 |
| Root Mean Square Error | 9.63492 |
| Mean of Response | 37.68774 |
| Observations (or Sum Wgts) | 4982 |

**Analysis of Variance**

| Source | DF | Sum of Squares | Mean Square | F Ratio |
|---|---|---|---|---|
| Model | 5 | 3332712.8 | 666543 | 7180.119 |
| Error | 4976 | 461930.4 | 93 | Prob > F |
| C. Total | 4981 | 3794643.3 | | <.0001* |

*Figure 22: Model 3 Summary of Fit and Analysis of Variance.*

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| |
|---|---|---|---|---|
| Intercept | 76.835847 | 2.86443 | 26.82 | <.0001* |
| social_media_hours | -8.075432 | 0.087 | -92.82 | <.0001* |
| study_hours | 9.7704574 | 0.069525 | 140.53 | <.0001* |
| sleep_hours | 4.9844441 | 0.099071 | 50.31 | <.0001* |
| notifications_per_day | -0.037595 | 0.001751 | -21.47 | <.0001* |
| focus_score | -0.746524 | 0.028659 | -26.05 | <.0001* |

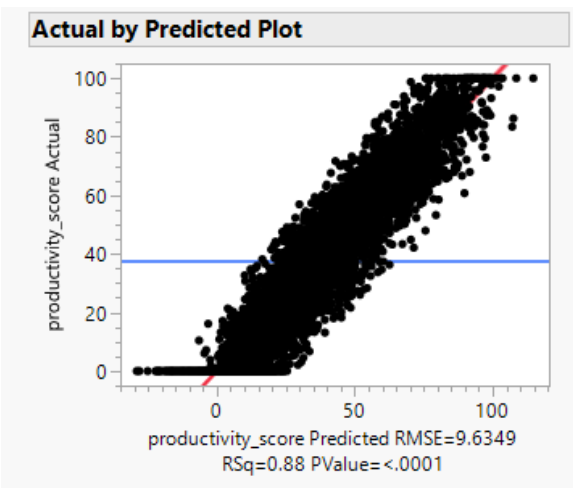



*Figure 23: Model 3 Parameter Estimates.* *Figure 24: Actual by Predicted plot.*

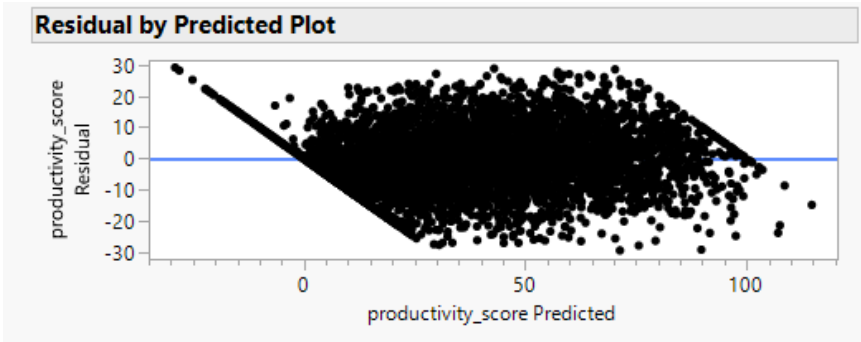



*Figure 25: Model 3 Residual by Predicted plot.*

Model 3 has R-squared 0.8783 and Adjusted R-squared 0.8781, identical to Model 2 to four decimal places, with an F-statistic of 7180.1 and a p-value below 0.0001. All five predictors are highly significant. The estimated coefficient on social_media_hours is -8.075, meaning each additional hour of daily social media use is associated with about an 8-point drop in productivity score, holding the other variables constant. study_hours has a coefficient of 9.77 (positive), sleep_hours 4.98 (positive), notifications_per_day -0.038, and focus_score -0.747. The negative coefficient on focus_score is again unintuitive given its positive correlation with productivity in the EDA, and motivates testing a quadratic term in the next model.

## Model 4: Adding Second-Order Terms

We tested second-order terms for each of the five remaining predictors by adding them one at a time to Model 3 and noting the p-value of the added quadratic. Two were highly significant: focus_score squared and study_hours squared (both $p < 0.0001$). The squared terms for sleep_hours, notifications_per_day, and social_media_hours were not individually significant. Model 4 adds focus_score squared and study_hours squared to Model 3.

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_5^2 + \beta_7 x_2^2$$

Where $x_5^2$ = focus_score squared and $x_2^2$ = study_hours squared.

**Hypothesis (for the addition):**

$$H_0: \beta_6 = \beta_7 = 0 \quad \text{vs.} \quad H_1: \text{at least one} \neq 0$$

**Output:**

**Summary of Fit**

| | |
|---|---|
| RSquare | 0.883709 |
| RSquare Adj | 0.883546 |
| Root Mean Square Error | 9.419004 |
| Mean of Response | 37.68774 |
| Observations (or Sum Wgts) | 4982 |

**Analysis of Variance**

| Source | DF | Sum of Squares | Mean Square | F Ratio |
|---|---|---|---|---|
| Model | 7 | 3353361.8 | 479052 | 5399.735 |
| Error | 4974 | 441281.5 | 89 | Prob > F |
| C. Total | 4981 | 3794643.3 | | <.0001* |

*Figure 26: Model 4 Summary of Fit and Analysis of Variance.*

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| |
|---|---|---|---|---|
| Intercept | 19.648075 | 4.81678 | 4.08 | <.0001* |
| social_media_hours | -7.691825 | 0.08869 | -86.73 | <.0001* |
| study_hours | 9.4977953 | 0.070425 | 134.86 | <.0001* |
| sleep_hours | 4.7941752 | 0.09765 | 49.10 | <.0001* |
| notifications_per_day | -0.034467 | 0.001724 | -19.99 | <.0001* |
| focus_score | -0.175431 | 0.048505 | -3.62 | 0.0003* |
| (study_hours-4.07688)*(study_hours-4.07688) | 0.2043494 | 0.028096 | 7.27 | <.0001* |
| (focus_score-96.3349)*(focus_score-96.3349) | 0.0241603 | 0.001781 | 13.57 | <.0001* |

*Figure 27: Model 4 Parameter Estimates.*

Model 4 yields R-squared 0.8837 and Adjusted R-squared 0.8835, an improvement of about half a percentage point over Model 3. The F-statistic is 5399.7. The estimated coefficient on study_hours squared is 0.204 ($p < 0.0001$), and on focus_score squared is 0.024 ($p < 0.0001$). Both second-order terms are positive, indicating that productivity increases at an accelerating rate as both focus_score and study_hours rise. With the quadratic terms in the model, the coefficient on focus_score becomes -0.175 and the coefficient on study_hours becomes 9.50, both still statistically significant.

**Partial F-Test: Model 3 vs. Model 4**

$H_0$: $\beta_6 = \beta_7 = 0$ (the coefficients on focus_score² and study_hours² are both zero)

$H_1$: at least one $\neq 0$

| | Sum of Squares (Error) | DF (Error) |
|---|---|---|
| Model 3 (reduced) | 461,930.3 | 4,976 |
| Model 4 (full) | 441,281.5 | 4,974 |

$$F = [(SSE_R - SSE_F) / (df_R - df_F)] / [SSE_F / df_F]$$

$$F = [(461{,}930.3 - 441{,}281.5) / 2] / [441{,}281.5 / 4{,}974]$$

$$\mathbf{F = 10{,}324.4 / 88.72 = 116.37, \quad df = (2, 4974), \quad p < 0.0001}$$

We reject the null hypothesis. The addition of focus_score squared and study_hours squared significantly improves the model.

## Model 5: Adding Interaction Terms

Building on Model 4, we tested several interaction terms motivated by the research question and the EDA. We tested social_media × study_hours, social_media × focus_score, social_media × sleep_hours, social_media × notifications_per_day, study_hours × focus_score, and study_hours × sleep_hours by adding each one at a time to Model 4 and noting the p-value. Two interactions stood out as both highly significant on their own and as providing the largest improvement to

Adjusted R-squared: social_media_hours × focus_score and study_hours × focus_score. Model 5 adds these two interactions to Model 4.

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_5^2 + \beta_7 x_2^2 + \beta_8(x_1 \cdot x_5) + \beta_9(x_2 \cdot x_5)$$

**Hypothesis (for the addition):**

$$H_0: \beta_8 = \beta_9 = 0 \quad \text{vs.} \quad H_1: \text{at least one} \neq 0$$

**Output:**

**Summary of Fit**

| | |
|---|---|
| RSquare | 0.887729 |
| RSquare Adj | 0.887526 |
| Root Mean Square Error | 9.256627 |
| Mean of Response | 37.68774 |
| Observations (or Sum Wgts) | 4982 |

**Analysis of Variance**

| Source | DF | Sum of Squares | Mean Square | F Ratio |
|---|---|---|---|---|
| Model | 9 | 3368616.7 | 374291 | 4368.211 |
| Error | 4972 | 426026.5 | 86 | Prob > F |
| C. Total | 4981 | 3794643.3 | | <.0001* |

*Figure 28: Model 5 Summary of Fit and Analysis of Variance.*

**Parameter Estimates**

| Term | Estimate | Std Error | t Ratio | Prob>\|t\| |
|---|---|---|---|---|
| Intercept | -2.266798 | 5.026152 | -0.45 | 0.6520 |
| social_media_hours | -7.312352 | 0.095083 | -76.90 | <.0001* |
| study_hours | 9.1910569 | 0.073154 | 125.64 | <.0001* |
| sleep_hours | 4.6540844 | 0.097221 | 47.87 | <.0001* |
| notifications_per_day | -0.031866 | 0.001717 | -18.56 | <.0001* |
| focus_score | 0.0539857 | 0.050794 | 1.06 | 0.2879 |
| (study_hours-4.07688)*(study_hours-4.07688) | 0.1342044 | 0.031745 | 4.23 | <.0001* |
| (focus_score-96.3349)*(focus_score-96.3349) | 0.0181165 | 0.001869 | 9.69 | <.0001* |
| (study_hours-4.07688)*(social_media_hours-4.14902) | -0.216535 | 0.037811 | -5.73 | <.0001* |
| (focus_score-96.3349)*(study_hours-4.07688) | 0.1001561 | 0.013968 | 7.17 | <.0001* |

*Figure 29: Model 5 Parameter Estimates.*

Model 5 reaches R-squared 0.8877 and Adjusted R-squared 0.8875, an additional improvement of about 0.4 percentage points over Model 4. The F-statistic is 4368.2. The interaction social_media_hours × focus_score has an estimated coefficient of -0.217 ($p < 0.0001$), indicating that the negative effect of social media on productivity is stronger when paired with high focus scores, or equivalently, the positive contribution of focus is dampened when paired with high social media use. The interaction study_hours × focus_score has a coefficient of 0.100 ($p < 0.0001$), indicating that high focus and high study hours together have a multiplicative positive effect on productivity beyond what each contributes alone. The main effect of focus_score becomes statistically insignificant in Model 5 ($p = 0.288$), which is expected when its interactions are included; we keep focus_score in the model for hierarchical consistency.

**Partial F-Test: Model 4 vs. Model 5**

$H_0$: $\beta_8 = \beta_9 = 0$ (the coefficients on social_media_hours × focus_score and study_hours × focus_score are both zero)

$H_1$: at least one $\neq 0$

| | Sum of Squares (Error) | DF (Error) |
|---|---|---|
| Model 4 (reduced) | 441,281.5 | 4,974 |
| Model 5 (full) | 426,026.5 | 4,972 |

$$F = [(SSE_R - SSE_F) / (df_R - df_F)] / [SSE_F / df_F]$$

$$F = [(441{,}281.5 - 426{,}026.5) / 2] / [426{,}026.5 / 4{,}972]$$

$$\mathbf{F = 7{,}627.5 / 85.69 = 89.01, \quad df = (2, 4972), \quad p < 0.0001}$$

We reject the null hypothesis. The addition of the two interactions significantly improves the model.

Having tested all main effects via stepwise selection, all viable second-order terms, and all viable interaction terms among the retained predictors, Model 5 represents the natural stopping point for our model-building process. No further additions improved fit without introducing instability among existing significant terms.

## Final Model and Validation

The final fitted model from JMP, with all coefficients reported in their centered form, is:

$$\hat{y} = -2.27 - 7.31 \cdot x_1 + 9.19 \cdot x_2 + 4.65 \cdot x_3 - 0.039 \cdot x_4 + 0.054 \cdot x_5$$

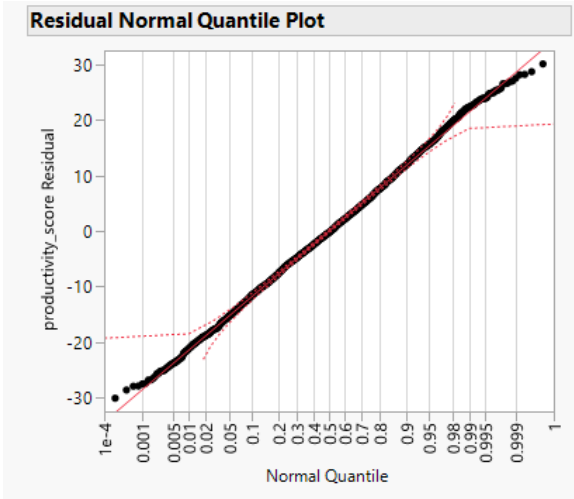
$$+ 0.018 \cdot (x_5 - 96.33)^2 + 0.134 \cdot (x_2 - 4.08)^2$$

$$- 0.217 \cdot (x_5 - 96.33) \cdot (x_1 - 4.15) + 0.100 \cdot (x_5 - 96.33) \cdot (x_2 - 4.08)$$

Where $x_1$ = social_media_hours, $x_2$ = study_hours, $x_3$ = sleep_hours, $x_4$ = notifications_per_day, and $x_5$ = focus_score. The squared and product terms are reported using the centered-and-scaled form that JMP applies by default in polynomial models.

Final fit statistics: R-squared 0.8877, Adjusted R-squared 0.8875, F-statistic 4368.2, RMSE 9.26, on 4,982 observations.

**Final Residual Diagnostics:**



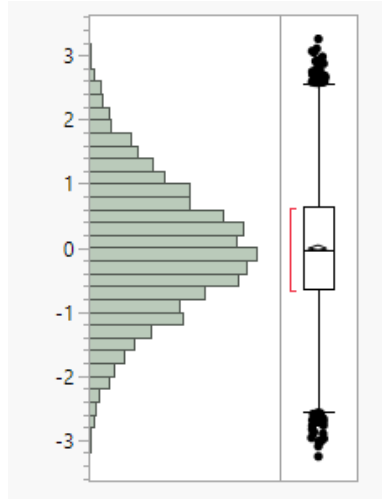
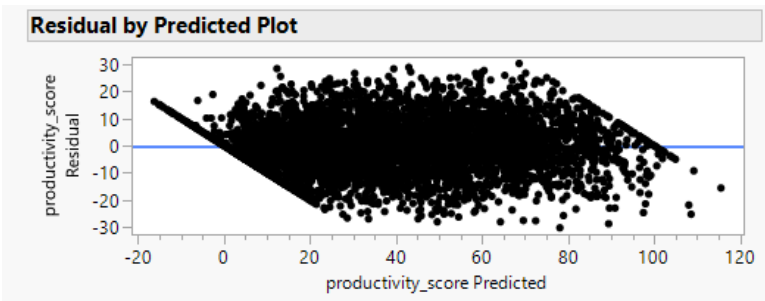




*Figure 30: Residual Normal Quantile Plot.* *Figure 31: Studentized Residuals distribution.*




**Durbin-Watson**

| Durbin-Watson | Number of Obs. | AutoCorrelation | Prob<DW |
|---|---|---|---|
| 2.0386741 | 4982 | -0.0193 | 0.9140 |

*Figure 32: Residual by Predicted plot.* *Figure 33: Durbin-Watson statistic.*

The residual versus predicted plot shows residuals scattered around zero with the constraint that some structure is visible at the lower tail; this reflects the floor effect noted in the EDA, where 14 percent of respondents scored exactly 0 on productivity. The diagonal pattern in the bottom-left of the plot is the expected appearance of points whose actual values are clipped at 0 but whose predicted values can extend below 0. Apart from this, residuals look unstructured and random.

The normal quantile plot of residuals is approximately linear through the bulk of the distribution, with mild deviation in the tails consistent with the boundary clustering. The studentized residuals histogram is roughly bell-shaped and symmetric around zero, with most observations falling within the conventional ±3 range. The Durbin-Watson statistic is 2.039 with autocorrelation -0.019 and p = 0.914, indicating no autocorrelation in the residuals.

**Validation:**

To check whether the model generalizes beyond the training data, we randomly split the cleaned dataset into approximately 80 percent training (3,968 observations) and 20 percent validation (1,014 observations) using JMP's Make Validation Column tool. The model was refit on the

training portion and used to predict productivity on the validation portion. Results are shown in Figure 34.

**Crossvalidation**

| Source | RSquare | RASE | Freq |
|---|---|---|---|
| Training Set | 0.8882 | 9.2394 | 3968 |
| Validation Set | 0.8858 | 9.2856 | 1014 |

*Figure 34: Crossvalidation results comparing training and validation sets.*

The training R-squared was 0.8882 with RASE 9.24, and the validation R-squared was 0.8858 with RASE 9.29. The negligible drop in performance between training and validation indicates that the model is not overfitting and generalizes well to data not used in fitting.

## Conclusions and Discussion

This analysis set out to test whether social media use specifically affects productivity beyond what total screen time can explain. The data supports that conclusion. In the first-order model that includes both daily screen time and daily social media hours as separate predictors, social media hours is highly significant with a coefficient of -8.06 ($p < 0.0001$), while total daily screen time is statistically insignificant with a coefficient of -0.02 and $p = 0.799$. The Effect Test for daily_screen_time in Model 1 confirms this directly: $F = 0.065$, $p = 0.799$. Once social media hours is in the model, raw screen time adds no useful information. Stepwise selection then dropped daily screen time entirely across all four selection methods, confirming that social media use, not screen time in general, is what carries the explanatory weight.

The final model identifies five core variables that predict productivity: social media hours, study hours, sleep hours, notifications per day, and focus score, with quadratic terms for focus and study hours and interactions between social media and focus, and study hours and focus. The model accounts for about 88.8 percent of the variance in productivity scores, which is a strong fit. The interaction between social media and focus score is particularly relevant to the research question; high social media use does not just lower productivity on its own, it also reduces the productive return that comes with high focus. The interaction between study hours and focus shows that focus and study time amplify each other; studying with high focus is more productive than the sum of either alone.

Self-reported addiction level was tested but added no explanatory power once actual hours were controlled for, suggesting that subjective self-classification tracks behavior less reliably than the measured behavior itself. Age was also non-significant, which is a useful null finding given the dataset spans ages 15 to 39.

The main caveats inherent to this analysis are tied to the data source. All variables are self-reported, so measurement error is likely. The data is cross-sectional, which means we cannot draw causal conclusions; the analysis describes association, not effect. The response variable

shows substantial clustering at the boundaries, with about 14 percent at 0 and a smaller share at 100, which may reflect either genuine responses or an artifact of how the data was generated, and which limits the usefulness of the model at the extremes. The Kaggle dataset was uploaded without published methodology or sampling details, which is itself worth noting. Further work could pair self-reported social media use with passively collected screen time data to reduce measurement error, and could use longitudinal data to assess how changes in social media use over time produce changes in productivity.

## Appendix

All relevant figures and JMP outputs included in main body.

## References

Zaman, A. Social Media Addiction vs Productivity Dataset. Kaggle. https://www.kaggle.com/datasets/asifxzaman/social-media-addiction-vs-productivity-dataset

JMP Statistical Discovery LLC. JMP Pro Software. SAS Institute Inc.